# YOUSSEF ASHRAF AWAAD EBRAHIM

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Military State: Exempted.

## SUMMARY

Entry-level Machine Learning Engineer with hands-on LLM evaluation and RLHF fine-tuning experience at Scale AI. Skilled in Computer Vision, NLP, and GenAI app development using LangChain and Azure OpenAI. Proficient in PyTorch, TensorFlow, and cloud platforms (Amazon SageMaker, Azure ML Studio), with a focus on building scalable, production-ready AI systems along with a special interest in Edge AI and resource-optimized deployments.

## PROFESSIONAL EXPERIENCE

**AI Engineer (LLM Evaluator) | Scale AI** (Project-based, remote) **08/ 2024 – 12/ 2024**

**Key Responsibilities:**

- **Conducted** evaluations for LLM-generated codes to align with **SWE-Bench**, focusing on real-world software issues collected from GitHub, via **RLHF fine-tuning** workflow.
- **Refined** models' outputs based on human preferences, **improving** the quality and coherence of their reasoning.
- **Evaluated** diverse prompt engineering techniques to **assess** LLM reliability and prompt adherence (instruction following, completeness, and accuracy) to improve model performance.
- **Collaborated** with distributed teams to refine training datasets, ensuring high-quality data for the models.

## TECHNICAL SKILLS

**Programming Languages:**

- Python - C (Embedded C) - C++ (OOP) - Java (JUnit Testing) - SQL

**Development Tools & IDEs:**

- Git & GitHub - Linux (WSL) - Anaconda - Jupyter Notebook - FastAPI - Postman
- VMware Player - Streamlit - Tkinter - MySQL Workbench - PostgreSQL - IntelliJ (JUnit5)

**AI Frameworks & Libraries:**

- Scikit-Learn - PyTorch - TensorFlow - Hugging Face Transformers - LangChain - Keras
- YOLO - OpenCV - SpaCy - NumPy - Pandas - Matplotlib - Seaborn

**Cloud ML Platforms & APIs:**

- Amazon SageMaker - Azure ML Studio - Gemini API - Google Search API - Azure OpenAI

## KEY PROJECTS

**AI-Powered Car Listing Assistant** | *Python + LangChain + Azure OpenAI + Streamlit UI.* **08 / 2025**

- **Engineered** a full-stack GenAI application for an online car sales platform, converting unstructured car listing data into structured JSON with Pydantic schemas, LangChain orchestration, and GPT-4o mini model.
- **Developed** a user-friendly interface enabling real-time image and text processing.
- **Designed** prompt engineering workflows with injection mitigation and token optimization for secure, cost-effective API use.

**Soccer Match Analyzer** | *Python + TensorFlow + OpenCV + YOLO.* **05 / 2025**

- **Developed** a computer vision system for analyzing soccer matches, focusing on player/ball detection.
- **Implemented** dynamic tactical analysis algorithms to map team formations and track players movement.

**EnGauge – AI Online Meeting Platform** | *Graduation Project (Grade: A+).* **07/ 2024**

A computer vision model that analyzes students' facial expressions and engagement level during online lectures.

**Key Responsibilities & Achievements:**

- **Architected** a CNN model based on ResNet50 for accurate engagement prediction.
- **Handled** large-scale preprocessing of 50+ hours videos data: DAiSEE and EngageNet datasets.
- **Resolved** the imbalanced class distribution problem in both datasets merged, **achieving a 12% improvement** in engagement prediction accuracy over existing work with the same datasets.
- **Generated** comprehensive reports about attendees' engagement levels, **providing** insights for the instructors.

**Tools & Technologies:** Python, TensorFlow, OpenCV, Kaggle Notebooks, and Microsoft Azure (ML).

**Image Classification App** | *Python + Deep Learning + CNN + PyTorch + ArgParse.* **10/ 2023**

- **Built** an Animals Image Classifier utilizing the "VGG16" pre-trained model for high-accuracy predictions.
- **Implemented** a command-line interface to facilitate model training and hyperparameter tuning.

## EDUCATION

**Bachelor's degree in computer and software engineering at Helwan University in Cairo** **09/ 2019 – 07/2024**

Cumulative Percentage: 80.45% (Very Good with Honors) | Cumulative GPA: 3.22/4.0

Relevant Coursework:

- *Statistics & Probability*
- *Data Structures & Algorithms*
- *Calculus & Linear Algebra*
- *Machine Learning*
- *Artificial Intelligence*
- *Software Engineering & Testing*
- *Database Management Systems*
- *Image Processing*

**STEM Egypt High School** **09/ 2016 – 07/ 2019**

Cumulative GPA = 3.7/4.0 | Achieved an A+ Grade in 5 Capstone Projects.

## CERTIFICATIONS & TRAINING

**AWS: Generative AI Scholarship** | *Foundations and Applications of LLM* **06/ 2025 – 07/ 2025**

Explored transformer architectures (e.g., BERT, GPT) and their applications in **NLP** tasks, applied prompt engineering, vector databases, and embeddings techniques, along with exploring **MLOps** with Amazon SageMaker.

**Udacity: Computer Vision Nanodegree** | *Advanced Computer Vision & CNN* **01/ 2025 – 04/ 2025**

Mastered image processing, **feature extraction**, **object detection**, and deep learning techniques with **CNNs** and **RNNs**, while practicing with **multi-object detection** models like **YOLO**.

**Siemens: Deep Dive Training Program** | *Developing ES with AUTOSAR Guide* **07/ 2023 – 09/ 2023**

AUTOSAR OS/Layers, COM Stack, MCAL modules, Adaptive topics, Advanced C++, and ISO26262.

**AWS: Deep Learning Scholarship** | *AI Programming with Python Nanodegree* **06/ 2023 – 10/ 2023**

Python Library Packages, Linear Algebra, Calculus, Neural Networks, and Deep Learning with PyTorch.

**National Telecommunication Institute: NTI** | *Artificial Intelligence (AI) Training* **07/ 2022 – 08/ 2022**

Python, Calculus and Statistics, Machine learning algorithms, and Deep learning techniques.

## EXTRACURRICULAR ACTIVITIES

**Hats Cookware Trading Event** | *Organizer Volunteer in an International Trading Summit.* **02/ 10/ 2022**

Organizing 15-minute one-to-one meetings for representatives from Egyptian Cookware factories and entities with investors, and importers from different countries across Africa, Europe, and Asia.

**Klexis Student Organization** | *Innovation Member & External Affairs Manager.* **01/ 2020 – 07/ 2021**

The external Affairs committee is responsible for Public Relations and Fundraising.

**STEM Emergency** | *Vice President & HR Head.* **07/ 2017 – 04/ 2018**

We built a website containing studying materials for helping STEM High school students prepare for exams and national competitions (e.g., Intel ISEF).

## LANGUAGES

- English: Upper Intermediate (B2 Level).
- Arabic: Mother Tongue.